MHD-WAAM THESIS | LITERATURE POSITION

The Empty Intersection

Three literatures on magnetised wire arc additive manufacturing, and the gap between them

10 September 2026 | search performed September 2026 | strict filter: arc-based additive manufacturing only

THE FINDING

Every study that measures the effect of a magnetic field on the microstructure of wire arc additive manufacturing is **experimental**. Not one is a simulation.

Simulation of WAAM microstructure **does** exist — cellular automaton grain prediction for nickel alloys, 921A steel and 316L — but **none of it includes a magnetic field**.

One simulation of magnetised WAAM exists (Zhou et al. 2024). It terminates at bead geometry and **never reaches grain structure**.

The three sets do not intersect. Every measurement of field-driven microstructural change in arc AM has been made without a model, and every model of arc AM microstructure has been built without a field.

This document reports a literature position and the gap that follows from it. Section 1 states the search method and its limits, because a claim that something does not exist is only as strong as the search that failed to find it. Sections 2 to 4 give the evidence. Sections 5 to 8 formulate the gap, the research question, the contribution and the terms on which the work can be falsified.

1. Method, and what it does not establish

Searches were run in September 2026 across the open literature for combinations of: wire arc additive manufacturing, WAAM, CMT, GMAW-based additive manufacturing, arc additive manufacturing; external magnetic field, alternating magnetic field, longitudinal magnetic field, electromagnetic stirring, magnetic arc oscillation; and microstructure, grain refinement, dendrite arm spacing, columnar to equiaxed transition. Simulation-side searches added cellular automaton, phase field, numerical simulation, magnetohydrodynamic and molten pool.

The filter applied

INCLUDED	EXCLUDED
Arc-based additive manufacturing: wire arc, CMT, GMAW-based, plasma arc, gas tungsten arc	**Laser processes.** Laser powder-bed fusion, laser beam welding, laser remelting, laser directed energy deposition
An externally applied magnetic field of any waveform or orientation	**Non-additive welding.** Single-pass welds that are not building a part
Microstructure reported: grain size or morphology, dendrite arm spacing, phase distribution, columnar-to-equiaxed transition	**Casting and directional solidification** under field, however similar the mechanism
Experimental or numerical, both admitted	**Field-free** WAAM microstructure work, and **microstructure-free** magnetic WAAM work — both are counted separately in Section 4

WHAT THIS SEARCH DOES NOT ESTABLISH

This is **not a systematic review**. It is a targeted search of the open literature accessible in September 2026, run over roughly a dozen query formulations. It cannot prove absence.

The honest formulation, and the one used throughout this document, is **"no study identified in this review"** rather than "no study exists". That phrasing should survive into the thesis.

Four papers listed in Section 2 are marked **unverified**: their titles report geometry or a confounded treatment, and full text was not obtained. They may or may not belong on the list, and the claim in Section 4 does not depend on them.

Publications after September 2026 are outside the search. The gap described here should be re-checked before submission.

2. Literature 1 — WAAM, magnetic field, microstructure: all experimental

These satisfy all three criteria. They are grouped by alloy system, with Al–5%Mg first because it is the alloy of this thesis.

2.1 Al–5%Mg

REFERENCE	FIELD	WHAT IS REPORTED
Zhao, Wei, Zhang, Chen & Ou (2022) *Int. J. Adv. Manuf. Technol.* 119:2571-2587 `10.1007/s00170-021-08466-5`	Longitudinal, **alternating**, square wave, 30-110 Hz, **4.6-13.5 mT** measured by gauss meter	Dendrite arm spacing **13 to 9 micron** at the bottom layer; **no change** mid-wall (32 vs 33 micron); pore diameter 80 to 31 micron; Al_3Mg_2 more uniform; tensile strength unchanged. Full text held
Effects of longitudinal alternating magnetic field on the microstructure and properties of CMT-WAAM Al-5%Mg alloy *Mater. Today Commun.* (2024) `PII S2352492824022530`	Longitudinal, alternating	Microstructure and properties. Title-confirmed; full text not obtained
Preparation and characterization of CMT-WAAM Al-5%Mg depositions through assisted longitudinal magnetic field (2023)	Longitudinal	Characterisation of deposits. Full text not obtained

2.2 Other aluminium alloys

REFERENCE	FIELD	WHAT IS REPORTED
Influence of assisted longitudinal magnetic field on the microstructure and mechanical properties of ER2319 aluminium WAAM *Sci. Technol. Weld. Join.* 28(3) (2023)	Longitudinal, assisted	Microstructure and mechanical properties. Title-confirmed
Droplet control and grain refinement in magnetic-assisted CMT additive manufacturing of 2319 aluminium *J. Mater. Res. Technol.* (2025)	Magnetic-assisted CMT	Droplet transfer control and grain refinement. Title-confirmed

2.3 Nickel superalloys, steel, titanium

REFERENCE	SYSTEM	WHAT IS REPORTED
Effect of magnetic field on the microstructure and mechanical properties of Inconel 625 WAAM *J. Manuf. Process.* 64:10-19 (2021)	Inconel 625	Microstructure and mechanical properties. Title-confirmed
Droplet transfer behaviour and microstructure of WAAM Inconel 718 assisted by an alternating magnetic field (2022)	Inconel 718	Droplet transfer and microstructure under an alternating field. Title-confirmed
Effect of electromagnetic stirring on microstructure and mechanical properties of AerMet100 ultra-high-strength steel WAAM *J. Mater. Res. Technol.* (2025)	AerMet100 steel	Microstructure and mechanical properties under electromagnetic stirring. Title-confirmed
Effects of synchronised magnetic arc oscillation on microstructure, texture, grain boundary and mechanical properties of WAAM Ti6Al4V *Additive Manufacturing* (2022)	Ti6Al4V	Microstructure, texture and grain boundary character. Title-confirmed

2.4 Listed but unverified

Four further papers combine WAAM with a magnetic field but report geometry, surface quality or a confounded treatment in their titles. Full text was not obtained and they are **not counted** in the argument of Section 4.

REFERENCE	WHY UNVERIFIED
Enhancing the geometry accuracy and mechanical properties of Al-5%Mg WAAM through longitudinal magnetic field influence, *J. Manuf. Process.* (2024)	Title reports geometry and mechanical properties; microstructure not stated. Same alloy and field as this thesis, so worth obtaining first
Preparation and characterization of high-efficiency deposition WAAM AlSi alloy assisted through magnetic field regulation and interlayer spraying lanthanum oxide powder, *J. Manuf. Process.* (2024)	**Confounded.** The field is applied together with a La_2O_3 inoculant, so any refinement cannot be attributed to the field alone
Enhancing wire arc additive manufacturing in vibrational environments with longitudinal magnetic fields, *Mater. Today Commun.* (2024)	Subject is mechanical vibration; the field is a remedy rather than the variable under study
Synergistic regulation of alternating magnetic field and torch tilt angle for self-supporting inclined components via CMT-WAAM, *J. Mater. Res. Technol.* (2026)	Subject is geometric precision of inclined walls; microstructure not stated in the title

One boundary case is set aside deliberately: **magnetic-field-assisted powder arc additive manufacturing of Ti60** (*Additive Manufacturing*, 2024) is arc-based and reports microstructure, but is powder-fed rather than wire-fed. Whether it falls inside the scope of "wire arc" is a judgement for the author.

THE COMMON PROPERTY OF SECTION 2

Every entry is experimental. Field applied, part built, sectioned, etched, imaged, measured. Not one of them contains a numerical model of any kind.

This is not a criticism. It is the state of the field, and it is precisely what makes the outcome robust: the *effect* has been established repeatedly, across four alloy systems and at least four process variants, by independent groups.

3. Literature 2 — WAAM microstructure simulation: all field-free

Predicting grain structure in wire arc additive manufacturing is an active and competent field. Cellular automaton methods coupled to thermal solvers are the dominant approach, and the arc-welding variant of the method (cellular automaton coupled to finite element, CAFE) has been established since at least 2014.

REFERENCE	SYSTEM	MAGNETIC FIELD?
Prediction of Ni-based alloy microstructure in wire arc additive manufacturing from a cellular automata model, *Comput. Mater. Sci.* (2023)	Ni-based alloy, WAAM	**None**
Multi-scale simulation of thermal processes and microstructure evolution in wire arc additive manufacturing of 921A steel (2025)	921A steel, WAAM	**None**
Grain morphology characteristics and formation mechanism of wire-arc additive manufactured 316L stainless steel (2025)	316L, WAAM	**None**
3D coupled cellular automaton / finite element modelling for solidification grain structures in gas tungsten arc welding, *ISIJ Int.* 54(2):401	GTAW, method paper	**None**
Three-dimensional cellular automaton / finite element modelling of solidification grain structures for arc-welding processes	arc welding, method paper	**None**

The methodological consequence is worth noting separately from the gap argument: **a route to grain structure in arc welding exists that does not require ExaCA**. CAFE for arc processes is established, which removes the risk carried since Phase 0 that no Al–Mg cellular-automaton response function exists.

4. Literature 3 — magnetised WAAM simulation: one paper, no microstructure

Reference	**Zhou, Fu, Zhou, Bai, Tian, Fu & Zhang (2024)**, "Numerical simulation of heat and mass transient behavior during WAAM overlapping deposition with external deflection magnetic field", *Int. J. Heat Mass Transfer* 218:124780. `10.1016/j.ijheatmasstransfer.2023.124780`
Field	External **deflection** magnetic field — longitudinal and deposition-direction components combined, not a pure longitudinal field
Method	Coupled arc and molten pool, by **weak coupling**: the arc is solved first and its output passed to the metal transport model as a fixed input. The droplet is a prescribed sphere at fixed frequency
Finding	The field produces circumferential and lateral force components, generating an **asymmetric rotating vortex** that drives melted metal into the gaps between overlapping beads and improves flatness
Subject	**Overlapping deposition** — bead-to-bead gaps in multi-pass walls, not single-bead solidification
Terminates at	**Bead geometry.** No grain structure, no dendrite arm spacing, no solidification criterion

One further entry belongs here: a preprint on numerical simulation of arc and metal transfer in arc-welding-based additive manufacturing under a longitudinal **static** field. It likewise stops short of microstructure.

5. The intersection

LITERATURE	MAGNETIC FIELD	MICROSTRUCTURE	MODEL
1. WAAM magnetic experiments (Section 2)	yes	yes	**no**
2. WAAM microstructure simulation (Section 3)	**no**	yes	yes

LITERATURE	MAGNETIC FIELD	MICROSTRUCTURE	MODEL
3. Magnetised WAAM simulation (Section 4)	yes	**no**	yes
The intersection	yes	yes	yes

THE INTERSECTION IS EMPTY

Every measurement of field-driven microstructural change in arc additive manufacturing has been made **without a model**.

Every model of arc additive manufacturing microstructure has been built **without a field**.

The one model of magnetised arc additive manufacturing **stops before microstructure**.

No study identified in this review occupies all three.

6. Why the answer cannot be borrowed from laser processes

The mechanism question — whether refinement comes from altered solidification conditions or from dendrite fragmentation — has been resolved, but only for laser processes. Ju et al. (2026, *npj Computational Materials*) couple thermoelectric magnetohydrodynamics to a dendrite-growth model containing fragment-induced nucleation, for AlSi10Mg under a static field in laser powder-bed fusion, and conclude that **fragmentation dominates**. Related phase-field work resolves AC-stirred dendrite growth in AA5754 during laser beam welding.

Those results are excluded from Sections 2 to 4 by the process filter, and deliberately so. But they must be addressed, because if their conclusion transferred, the gap would be of method rather than of knowledge. Three differences argue that it may not transfer.

	LASER POWDER-BED FUSION	ARC DEPOSITION
Field-free baseline	No impulsive agitation. The field **creates** the shear that breaks dendrite arms	A droplet strikes the pool at 0.9 m/s every 33 ms, giving a local shear rate of about **8×10^2 s^{-1}**, roughly **four times** the ambient pool shear. **Fragmentation already occurs without any field**
Shear regime	Pool ~100 micron at 1-5 m/s: **10^4 to 5×10^4 s^{-1}**	Pool ~5 mm at 0.3-1.6 m/s (measured, this project): **60 to 320 s^{-1}**. Fifty to several hundred times lower
Forcing	Static field. Monotonic loading of a dendrite arm	Alternating at 70 Hz — **4.7 reversals between consecutive droplets**. Cyclic loading is a different failure mode
Current source	Thermoelectric, driven by the interfacial temperature gradient	**Injected** welding current, 135 A at about 10^7 A/m^2. A different spatial distribution of J, and therefore of J x B

If droplet impingement already saturates fragmentation in arc deposition, the field's marginal contribution through that path could be negligible and the observed refinement would have to be thermal — the opposite of the laser result. The question is therefore open on physical grounds, not merely unasked.

7. The gap, formulated

The following is written as thesis prose and may be adapted directly. The hedged phrasing is deliberate and should be preserved.

THE GAP

What is established. Applying an external magnetic field during arc-based additive manufacturing refines the solidification microstructure of aluminium alloys. Zhao et al. (2022) measure secondary dendrite arm spacing falling from 13 to 9 micron in the first layer of Al-5%Mg under a 10.4 mT longitudinal alternating field, and the effect is reproduced across further Al-5%Mg studies, ER2319 and 2319 aluminium, Inconel 625 and 718, AerMet100 steel and Ti6Al4V. The existence of the effect is not in question.

What is contested is the mechanism. Two are proposed and they operate simultaneously. The thermal-convective path holds that the field dilates the arc and homogenises pool temperature, flattening the thermal gradient at the solidification front and favouring the columnar-to-equiaxed transition. The mechanical path holds that stirring shears primary dendrite arms until they detach, and that the fragments survive as heterogeneous nuclei. Zhao invokes both and, in common with every experimental study in this area, cannot separate them: a single act switches both on, and both produce the same signature of a finer, more equiaxed structure.

The separation is inaccessible to experiment and requires a model in which one mechanism can be suppressed. No such model exists for arc-based additive manufacturing. Simulation of magnetised arc deposition terminates at bead geometry; simulation of arc deposition microstructure omits the field; and the mechanism question has been resolved only for laser processes, in a regime whose field-free baseline, shear rate and forcing waveform all differ from arc deposition by amounts that could reverse the conclusion.

Research question

Does field-driven grain refinement in arc-based additive manufacturing operate by the same mechanism as in laser powder-bed fusion?

Specifically: what fraction of the refinement observed under a longitudinal alternating field in Al-5%Mg WAAM is attributable to altered solidification conditions, and what fraction requires dendrite fragmentation?

Contributions

#	CONTRIBUTION	WHY IT IS NEW
1	**Welding-current injection into an open-source magnetohydrodynamic VOF solver**	The magnetic module of the chosen solver carries only motional current, sigma(u x B), two to three orders of magnitude below the arc regime. Injecting the arc current, validated against published analytic force fields at zero applied field, is a prerequisite that does not exist in open form
2	**Quantification of G, R and cooling rate at the solidification front in magnetised arc deposition**	Neither the thermal gradient nor the front velocity can be measured inside a millimetric pool at 1500 K. They are the inputs every solidification criterion requires, and no study identified in this review reports them under an applied field for an arc process

#	CONTRIBUTION	WHY IT IS NEW
3	**Decomposition of the bead-geometry response into arc-mediated and pool-mediated contributions**	Obtained by suppressing each in turn, which experiment cannot do. Zhou et al. compute both but do not separate them
4	**A test of regime-dependence for the laser-process conclusion**	Ju et al. (2026) find fragmentation dominant in laser powder-bed fusion. Whether that survives a fifty-fold larger pool, a pre-existing droplet-driven shear source and a reversing rather than steady stir has not been asked

8. Falsifiability, limitations, and what remains to be checked

8.1 The prediction, made before the run

Zhao's measured refinement from 13 to 9 micron implies, through the coarsening relation between secondary dendrite arm spacing and cooling rate,

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lambda_2 ~ (G x R)^(-1/3) -> (13/9)^3 = 3.0
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a **three-fold** increase in local cooling rate if altered solidification conditions alone are responsible. The model computes that increase directly, and the outcome decides between two answers.

IF THE MODEL COMPUTES	THEN
Cooling rate rises by about **3x**	The thermal mechanism is sufficient. Fragmentation is unnecessary in arc deposition, and the laser-regime conclusion of Ju et al. is **regime-specific**
Cooling rate rises **substantially less**	Fragmentation is required in arc deposition also. The laser conclusion **generalises**, and the shortfall quantifies the fragmentation contribution

Both outcomes are informative, and the decision rule is fixed in advance.

8.2 Declared limitations

TO BE STATED IN CHAPTER 1, NOT BURIED

Fragmentation is not modelled explicitly. Its contribution is obtained by difference rather than computed, and inherits the uncertainty in the coarsening exponent — which over the plausible range 0.30 to 0.40 moves the required cooling-rate ratio from 3.4 to 2.5.

A fixed Goldak source cannot dilate under field. The arc-mediated share is therefore imposed empirically, calibrated against measured bead cross-sections, rather than predicted from first principles.

The droplet is injected with prescribed mass, temperature and momentum. Effects of the field on droplet transfer — which Zhao reports, and which his own data show to be substantial — are outside scope.

Stating these in the first chapter converts them from vulnerabilities into scope. The second and third are what a follow-on project would address.

8.3 Open actions on the review itself

ACTION	WHY
Obtain full text of *Enhancing the geometry accuracy and mechanical properties of Al-5%Mg WAAM through longitudinal magnetic field influence* (2024)	Same alloy, same field orientation as this thesis. If it reports microstructure it belongs in Section 2.1 and is the nearest neighbour to the present work

ACTION	WHY
Obtain full text of Ju et al. (2026)	If their dendrite-fragment nucleation model is portable, the fragmentation contribution can be **computed** rather than inferred by difference, which would remove the principal limitation in 8.2
Obtain the AA5754 phase-field paper (IJHMT 2023)	AC-stirred dendrite growth in an Al-Mg alloy. Closest available methodological template for the microstructure stage
Re-run this search before submission	The field is active: three of the Al-5%Mg experimental papers post-date the 2022 source paper, and one candidate is dated 2026

A final note on framing. The project description as originally written — analysing the effect of a magnetic field during WAAM and observing microstructure — describes work that has now been done at least seven times. The reformulated question in Section 7 describes the same work and claims something nobody has established. The difference is not cosmetic: it is the difference between a replication and a result.